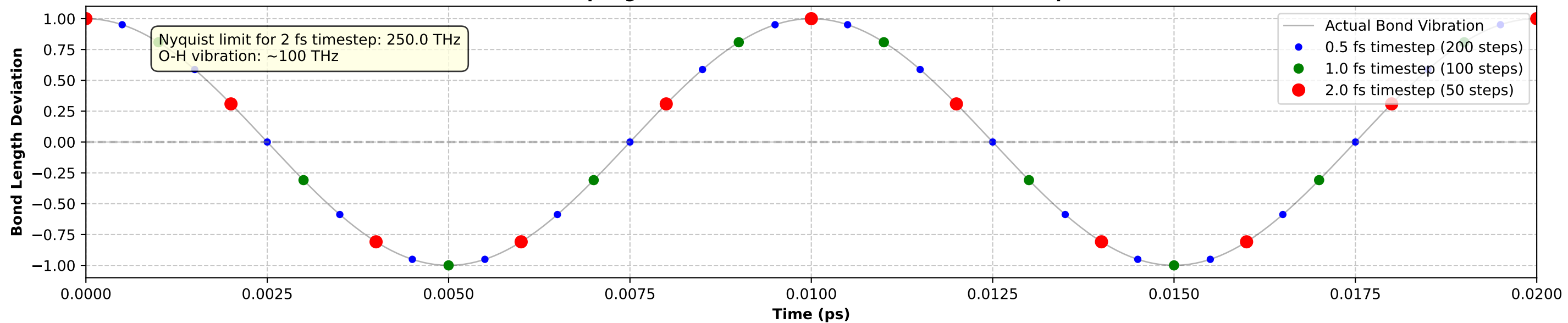
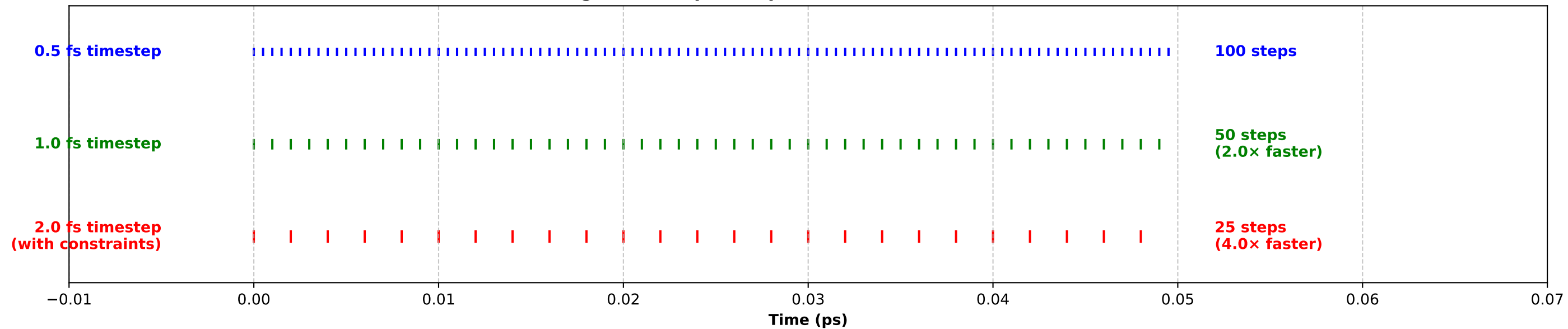


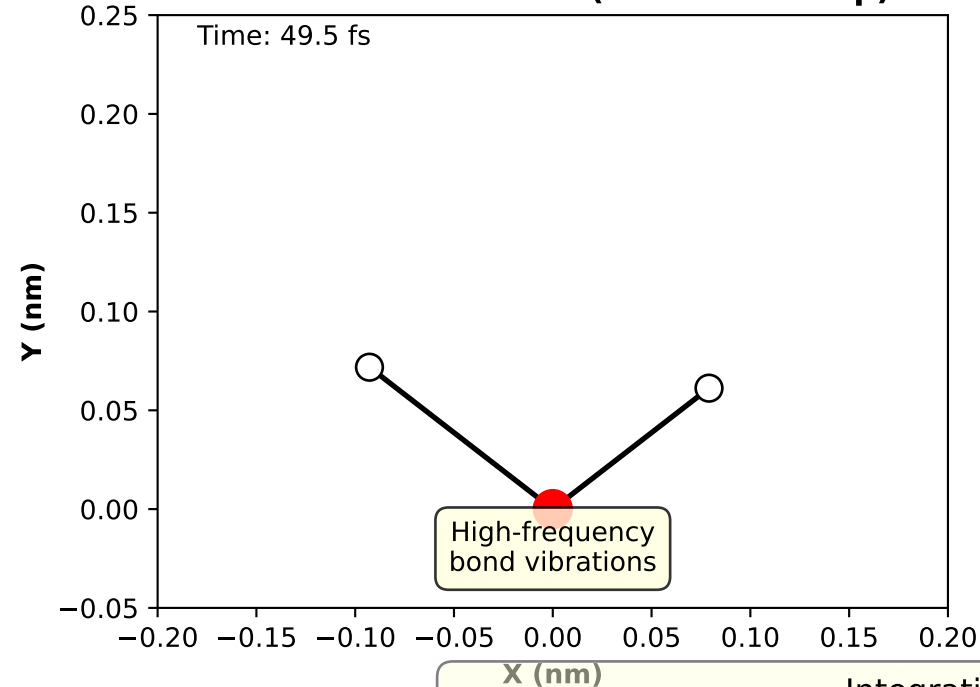
Sampling of Bond Vibration with Different Time Steps



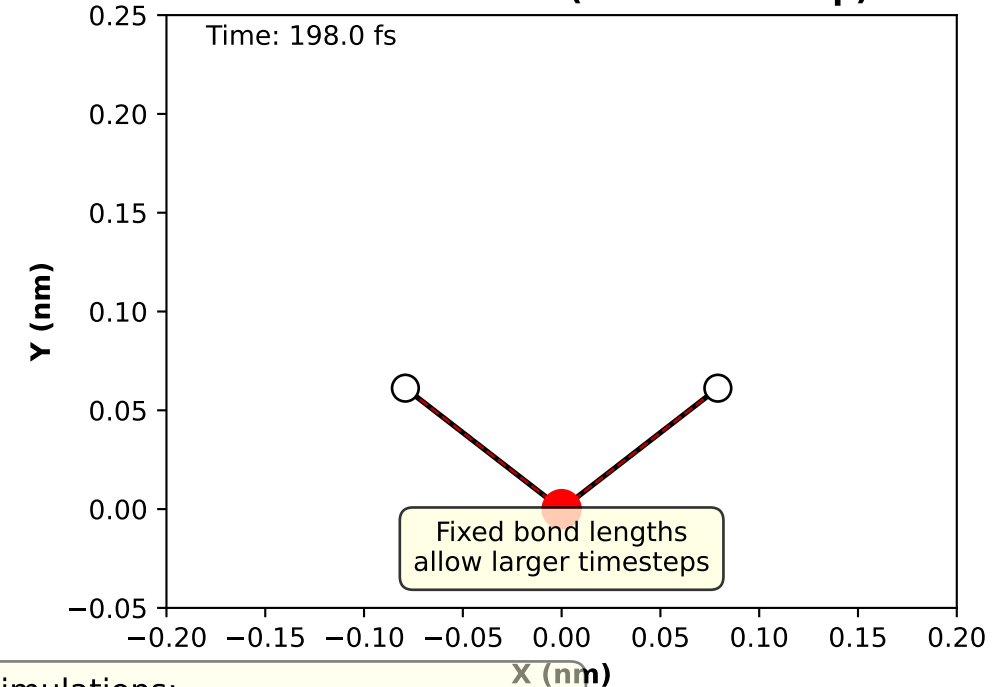
Integration Steps Comparison for 50 fs of Simulation



Without Constraints (0.5 fs timestep)



With Constraints (2.0 fs timestep)



Integration Time Step in MD Simulations:

- Without constraints: Bond vibrations limit time step to ~0.5-1.0 fs
- With constraints (SHAKE/LINCS): Fixed bond lengths eliminate high-frequency vibrations
 - Larger time step (2.0 fs) possible with constraints, making simulations 2-4× faster
 - Constraints preserve the essential physics while improving computational efficiency